



Experimental investigation on the vortex-induced vibration of an arch steel bridge tower

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ABSTRACT

This paper reports the wind tunnel study of the vortex-induced vibration (VIV) of a free-standing steel arch bridge tower with pentagon cross-section. The tests were carried out at different yaw angles β (0° – 90°) of wind flow by employing a 1:50 scaled aeroelastic model. A series of aerodynamic measures were studied to mitigate the VIV of the tower at the most unfavorable yaw angle. Effects of turbulence intensity and structural damping on the VIV performance of the tower were evaluated. A sleeve-type eddy current damper (ECD) for increasing the structural damping of aeroelastic model was designed for reserved emergency measures. Experimental results indicate that the VIV occurs for a few yaw angles at wind velocity of 15–18 m/s and the maximum VIV amplitude happens for $\beta = 35^\circ$. The aerodynamic measures are effective for some yaw angles but become invalid for other yaw angles. Compared to those in smooth flow, the VIV amplitude drops by 65% for a turbulence intensity of 8.5%. The VIV of the tower can be effectively suppressed by the eddy current damper which provides additional damping to the aeroelastic model. The sleeve-type eddy current damper is shown to be a convenient device to adjust the damping for aeroelastic models.

1. Introduction

Long-span cable-supported bridges are very sensitive to wind loads due to their flexibility and lightness and low damping ratios. Especially, during construction phase, bridge towers of cable-stayed or suspension bridges can be very sensitive to wind loads as these free-standing towers are not restrained by stay cables and often exposed to wind in an open terrain with low turbulence intensity (Larose et al., 1998). Meanwhile, vortex-induced vibration (VIV) is more likely to happen for steel bridge towers due to lower stiffness and damping (Siringoringo and Fujino, 2012; Fujino, 2002). VIV significantly affects the safety of construction workers and tower structures. Therefore, it is necessary to study the VIV characteristics of a new type bridge tower and propose efficient VIV mitigation measures.

The VIV characteristics of a bridge tower are affected by its geometrical properties which depend on the cross-section type and bridge tower type. Usually, bridge tower cross-sections are rectangular or other aerodynamically bluff sections. Matsumoto et al. (1998) studied the effects of the yaw angles (0° – 90°) of the inflow wind on the aerodynamic performance of a rectangular tower through wind tunnel tests using two- and three-dimensional models. It was found that both the

two- and three-dimensional rectangular tower models are very sensitive to yaw angles, and the amplitude of the VIV in the cross-wind direction reaches its maximum when the yaw angles are 0° and 90° . In addition, according to the wind tunnel test, results showed that the VIV characteristics of bridge towers depend on the spacing ratio between two cylinders of bridge tower (Takeuchi, 1990). Aerodynamic interference effect is dependent on the spacing ratio, which is affected by the type of bridge towers. The VIV characteristics of bridge towers are also influenced by different bridge tower types such as single column, H-shape, A-shape, inverted Y-shape and masonry-shape, etc. Zhu et al. (2011) carried out wind tunnel tests with an aeroelastic model to study the VIV characteristic of a curve steel bridge tower in free-standing state with lower stiffness and damping. Ma et al. (2019) estimated the aeroelastic response characteristics of a single-column bridge tower with a chamfered square cross-section which was exposed to a smooth flow in the wind tunnel tests. Larose et al. (1995) experimentally investigated the VIV responses of the tower top for a suspension bridge under free-standing state, and the effect of smooth inflow wind (with yaw angles 0° , 5° , 10° , 15° , 20° and 25°) and turbulent inflow wind (with yaw angles 0° , 30° and 80°) were studied.

The VIV characteristics of single-column bridge towers and four-

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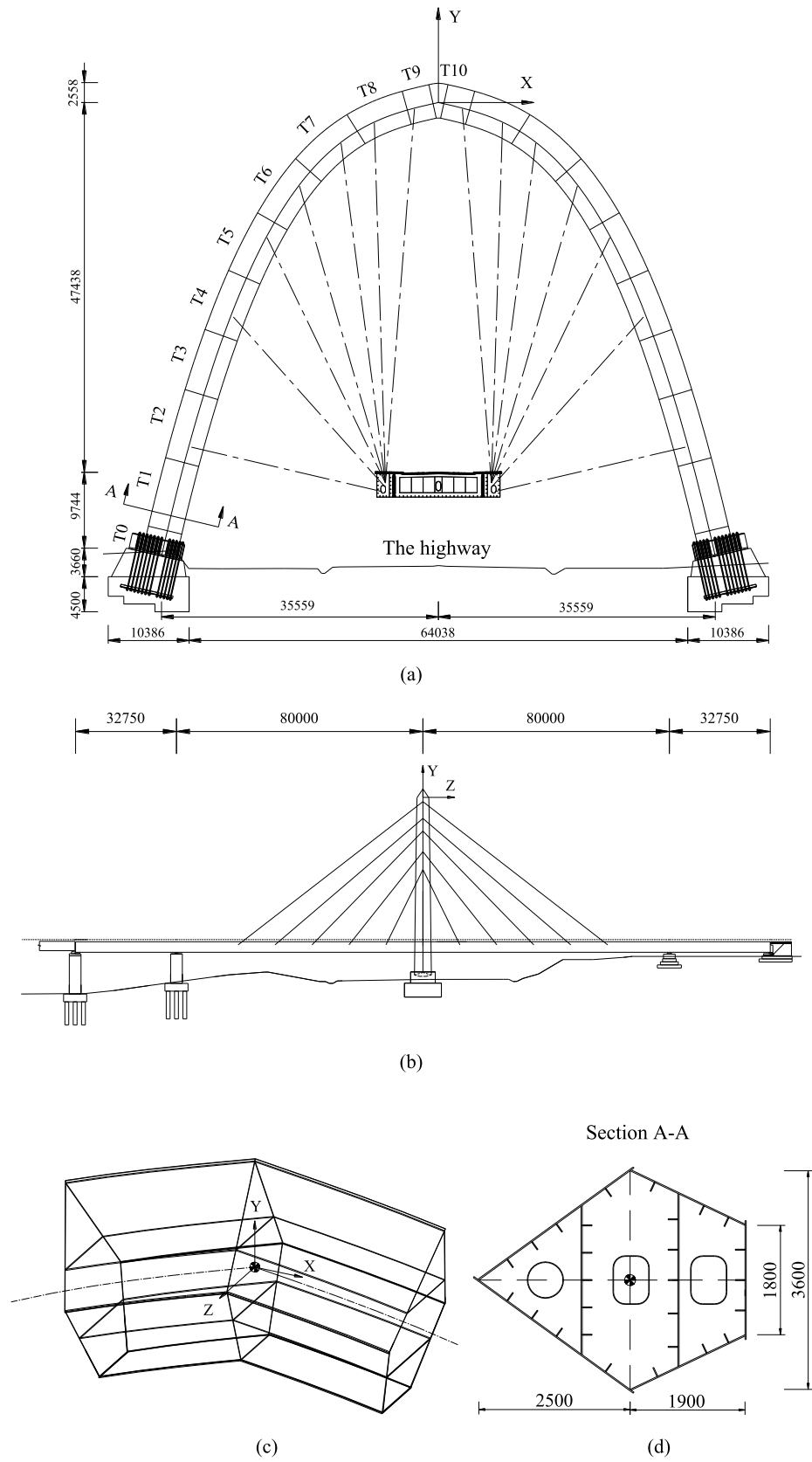


Fig. 1. Cable-stayed bridge with arch steel tower (unit: mm): (a) side view of the bridge; (b) elevation view of the bridge; (c) isometric view of tower segment; (d) typical cross-section of bridge tower.

Table 1

The cross-sections of the arch steel bridge tower.

Cross-section	Thickness of outer plate (mm)	Thickness of web(mm)	Thickness of stiffening rib (mm)	Thickness of diaphragm plate (mm)
T0~T1	28	28	24	20
T2	28	28	20	20
T3	24	24	20	20
T4~T10	20	20	16	16

column bridge towers have been experimentally and numerically studied (Nguyen et al., 2018; Marra et al., 2017; Belloli et al., 2011; Matsumoto et al., 2008; Fang et al., 2020). Furthermore, wind-induced vibrations of bridge towers were observed by in-site experiments, and it was found that the vortex shedding is caused by the fact that the vibration of the bridge tower with a bluff body cross section is dominated by oscillations at a single frequency (Siringoringo and Fujino, 2012; Larose et al., 1998). These studies showed that the bridge towers' VIV characteristics are very important, especially for steel bridge towers whose VIV phenomenon can be very hazardous. Meanwhile, it is necessary to study a number of mitigation measures to avoid excessive VIV amplitude in the next section, including aerodynamic measures and damping measures.

As far as rectangular section, corner cut of bridge tower cross-sections was a common aerodynamic measure to improve the aerodynamical performance (Takeuchi, 1990; Li et al., 2012; Shiraishi et al., 1988; Ogawa et al., 1990). Based on wind tunnel tests, the porous screens attached to the edges of the bridge tower cross-section shape are effective in mitigating the suspension bridge tower's VIV (Larose et al., 1995). The VIV performance of the curve steel bridge tower can be improved by the aerodynamic mitigation measures including flow-disturbing rails, multi-orifice angle-shape flow disturbers and multi-orifice flow-disturbing plates. The multi-orifice flow-disturbing plate has a better mitigation ratio than the other aerodynamic mitigation measures, and the VIV mitigation ratio is 71% (Zhu et al., 2011). The VIV phenomenon was observed for a 1:80 scaled aeroelastic model, and curve guide vanes and vertical fins were found able to effectively reduce the VIV amplitude. The curve vanes were found generally more effective (Ma et al., 2019). Additionally, damping devices can be used to effectively suppress the wind-induced vibration of bridge towers in construction, such as active tuned mass dampers, magnetorheological elastomers variable stiffness tuned mass dampers, tuned liquid dampers and multi-stage pendulum mass dampers (Inoue et al., 2014; Wang et al., 2020; Fujino, 2002; Guo et al., 2007a,b). In order to suppress the vibration to an allowable level, tuned mass dampers and pendulum tuned mass dampers were installed in a suspension bridge tower as the effect of aerodynamic measures is insufficient (Kitagawa, 2004). VIV mitigation effects of electromagnetic dampers on the cable-stayed bridge tower's aeroelastic model were experimentally investigated (Ogawa et al.,

1990). Effects of tuned mass dampers on reducing the VIV amplitude of the suspension bridge tower with a smooth flow or a turbulent flow were studied (Ge et al., 2007). In conclusion, mitigation measures are important to improve the bridge towers' VIV performance during construction phase, as the bridge towers' VIV has an impact on the construction quality and security of constructors. Therefore, it is valuable to investigate the VIV characteristics and mitigation measures of bridge towers.

Despite that VIV has already been studied for many types of bridges towers in the literature, the VIV characteristics of arch bridge towers were rarely reported. Moreover, the effect of sleeve-type eddy current damper (A simple device that provides adjustable additional damping to the aeroelastic model by changing the number of magnets) on mitigating the arch bridge tower's VIV has never been experimentally studied. This paper aims at experimentally studying the VIV characteristics of an arch steel bridge tower in a stayed-cable bridge shown in Fig. 1. Table 1 lists the geometric properties of the cross sections of the arch steel bridge tower. In order to investigate the steel bridge tower's VIV characteristics, a 1:50 scaled aeroelastic model was designed for the wind tunnel tests. These tests were carried out in HNU-2 wind tunnel in Hunan University. The effect of yaw angles on the VIV characteristics of the arch bridge tower in the smooth flow was studied. Besides, to reduce the VIV amplitude, several mitigation measures including aerodynamic mitigation measures and installation of a sleeve-type eddy current damper were proposed and investigated. Meanwhile, to analyze the mechanism of the aerodynamic mitigation measures, the VIV performance of the arch steel bridge tower with respect to turbulence intensity was studied.

2. Aeroelastic model

2.1. Dynamic properties

The aeroelastic model of the bridge tower in this study is based on a cable-stayed bridge. The span of the cable-stayed bridge is 225.5m, and the height and span of the arch steel bridge tower are 59.74m and 71.118m, respectively, as shown in Fig. 1. A coordinate system is defined such that the global X, Y and Z directions (see Fig. 1) are the longitudinal, vertical and lateral of the bridge respectively. The isometric view and typical cross-sections of the steel arch bridge tower are given in Fig. 1(c) and 1(d). The cross-section of the tower is a pentagonal steel box with variable thicknesses.

The design of the aeroelastic model requires the dynamic characteristics of the arch bridge tower in its free-standing state. The dynamic properties of the free-standing tower were obtained from a finite element model established in ANSYS as shown in Fig. 2. In the finite element model, three-dimensional beam element 'BEAM188' is used to simulate the bridge tower, and the additional mass of the diaphragm plate is modelled by MASS21 element. The bottom of the arch bridge

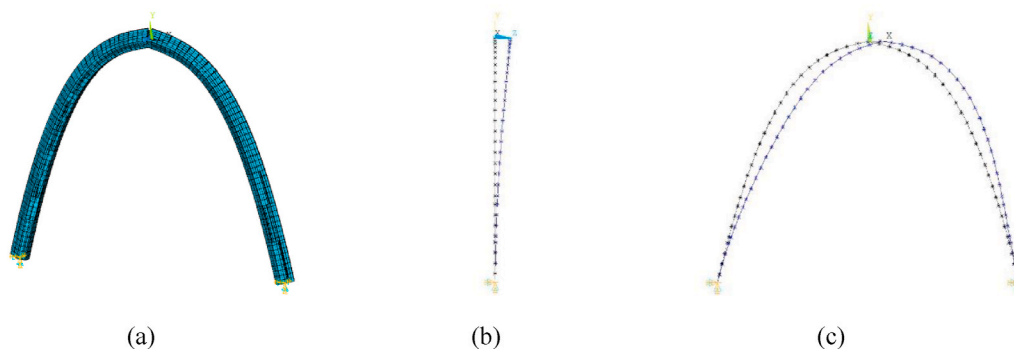


Fig. 2. FE model and dynamic properties of the free-standing arch steel tower: (a) FE model; (b) out-of-plane vibration mode (0.694Hz); (c) in-plane vibration mode (1.994Hz).

Table 2

Dynamic property of free-standing bridge tower.

Mode order	Frequency (Hz)	Mode shape
1	0.694	The first longitudinal bending (out-of-plane)
2	1.994	The first lateral bending (in-plane)
3	2.016	The first torsion mode
4	3.960	The first symmetric torsion mode
5	4.841	The first symmetric vertical bending

Table 3

Scaling ratios for the aeroelastic model.

Parameter	Symbol (unit)	Scaling ratio
Length	L (m)	1:50
Velocity	U (m/s)	1:7.07
Air density	ρ (kg/m ³)	1:1
Frequency	f (Hz)	7.07:1
Mass per unit length	m (kg/m)	1:502
Bending rigidity	EI (N·m ²)	1: 505
Damping ratio	ζ (none)	1:1

Table 4

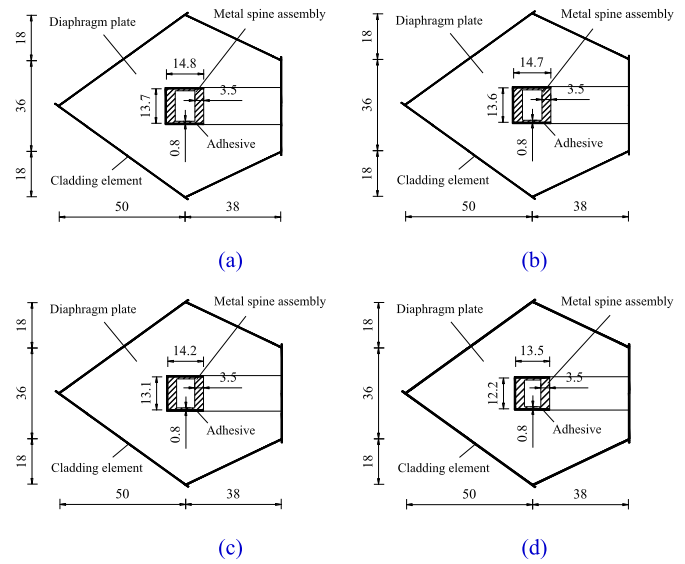
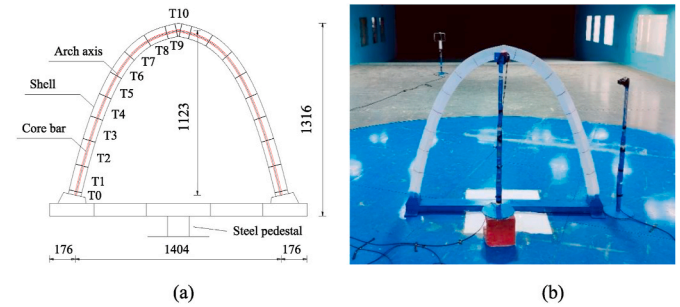
Cross-sectional bending rigidities of prototype and aeroelastic model.

Cross-section	Bending rigidity	Prototype (N·m ²)	Model (N·m ²)		Discrepancy (%)
			Target	Designed	
T0~T1	EI_x	1.3274×10^{11}	424.8	424.2	-0.14
	EI_z	2.1050×10^{11}	673.6	676.7	0.46
T2	EI_x	1.2934×10^{11}	413.9	414.3	0.10
	EI_z	2.0410×10^{11}	653.1	660.1	1.07
T3	EI_x	1.1577×10^{11}	370.5	367.0	-0.94
	EI_z	1.8216×10^{11}	582.9	581.3	-0.27
T4~T10	EI_x	0.9297×10^{11}	297.5	293.5	-1.34
	EI_z	1.4709×10^{11}	470.7	474.3	0.76

tower is fixed, with the translations and rotations in all directions constrained. The fundamental mode of the free-standing tower is its out-of-plane vibration (along weak axis), and the second-order mode of the tower is its in-plane vibration (along strong axis), as shown in Fig. 2(b) and 2(c). The frequencies of these two modes are 0.694Hz and 1.994Hz, respectively. Table 2 lists the first five modes in the frequency range 0–5Hz. The frequency for the first-order torsion mode is 2.016Hz which is very close to the first-order lateral bending frequency only with a difference of 0.022Hz. The box cross section of the bridge tower has considerable torsional resistance, so torsional vibration is not considered in this study.

2.2. Scaling rules of the aeroelastic model

A 1:50 scaled aeroelastic model of the arch steel bridge tower is designed to fit for the 8.5m × 2m test section in the HD-2 wind tunnel. In addition to the geometric similarity, similarity requirements of the mass distribution, stiffness and damping characteristics for the aeroelastic model and the prototype should also be satisfied, as shown in Table 3. The aeroelastic model is composed of a metal spine assembly inside to provide the stiffness of the bridge tower, and cladding elements to simulate its aerodynamic shape. The additional mass was carefully configured to obtain a correct mass distribution of the aeroelastic model.

**Fig. 3.** Cross-sections of the aeroelastic model (unit: mm): (a) T0~T1; (b) T2; (c) T3; (d) T4~T10.**Fig. 4.** Aeroelastic model of the arch steel bridge tower (unit: mm): (a) design; (b) completed.

2.3. Aeroelastic model design

The dimensions of the metal spine assembly were chosen to assure its longitudinal (out-of-plane direction) bending stiffness and lateral (in-plane direction) bending stiffness are correctly scaled, while the cladding elements were designed in a full accordance to the aerodynamic shape of cross-sections of the arch tower.

In order to make the metal spine assembly meet the similarity requirements of the longitudinal and lateral bending stiffness simultaneously, A3 steel was selected for the metal spine assembly, and the cross section of the metal spine assembly was selected as a box section. The target and designed moments of inertia of the cross sections about different axes for the metal spine assembly are listed in Table 4, compared to those for the prototype. It is shown that the designed bending rigidities of the metal spine assembly are very close to the target values, and the maximum discrepancy is -1.34%.

The aeroelastic model of the bridge tower was fabricated with the parameters designed above, with its cross sections shown in Fig. 3. The metal spine assembly is glued to the diaphragm plate by adhesive, and then the cladding element is glued to the diaphragm plate. The mass of the prototype is 627.007tons, corresponding to a target mass of 5.016 kg for the aeroelastic model. As the mass of the metal spine assembly is 2.919 kg, an additional mass of 2.097 kg should be added. Lead blocks were used as the additional mass, and the lead blocks are glued to the metal spine assembly. The cladding element is made of acrylonitrile butadiene styrene (ABS) material. The aeroelastic model of the arch steel bridge tower is shown in Fig. 4. The designed and actual aeroelastic



Fig. 5. Details of the aeroelastic model: (a) installation of steel pedestal; (b) displacement measurement.

Table 5
Dynamic characteristics of prototype and aeroelastic model.

Mode of vibration	Frequency of prototype (Hz)	Frequency of model (Hz)		Discrepancy (%)	Damping ratio (%)
		Target	Measured		
Longitudinal bending mode	0.694	4.907	4.872	-0.71	0.30
Lateral bending mode	1.994	14.100	13.624	-3.38	0.55

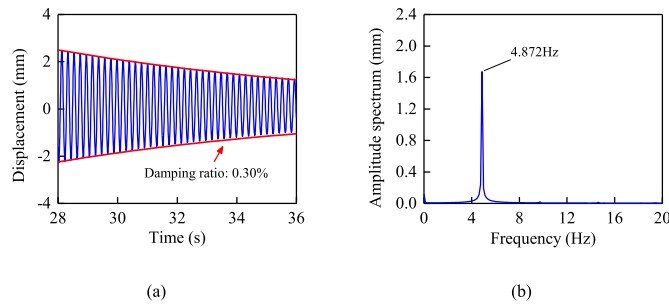


Fig. 6. Longitudinal displacement of the aeroelastic model from free decay tests: (a) Longitudinal displacement; (b) power spectrum.

models are shown in Fig. 4(a) and 4(b) respectively. The steel pedestal is bolted to a turntable support platform to provide sufficient bottom stiffness, as shown in Fig. 5(a). The marked points for displacement measurement are located on the top and at the two sides of the bridge tower, respectively, as shown in Fig. 5(b).

2.4. Dynamic characteristics test of aeroelastic model

To ensure that the aeroelastic model can truly reflect the dynamic characteristics of the prototype, dynamic characteristics tests of the aeroelastic model were carried out. The bending modes in the longitudinal and lateral directions were artificially excited considering the different vibration characteristics of these modes. The displacement signals were measured by two laser displacement meters, and the test data were collected and stored by DH5922D data acquisition system. The sampling frequency is 200Hz during the tests. The natural frequencies and damping ratios of the prototype and aeroelastic model are listed in Table 5. According to the test results, the target and measured frequencies of the aeroelastic model for the bending mode in the longitudinal direction are 4.907Hz and 4.872Hz respectively, while the expected and measured frequencies for the bending mode in the lateral direction are 14.100Hz and 13.624Hz respectively. The discrepancies between the target and measured frequencies for the aeroelastic model

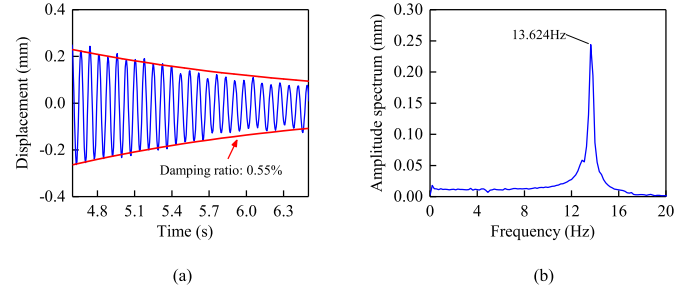


Fig. 7. Lateral displacement of the aeroelastic model from free decay tests: (a) Lateral displacement; (b) power spectrum.

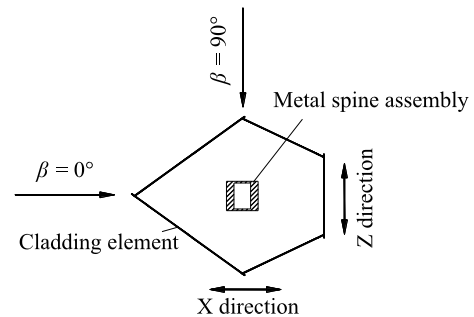


Fig. 8. Definition of the yaw angle β .

are less than 3.5%, which is satisfactory according to design specification by (MOT, 2018). The discrepancies could be caused by fabrication and installation processes. The damping ratios were calculated using the logarithmic decrement method on the data obtained from the free decay tests. For the longitudinal and lateral modes, the damping ratios are 0.30% and 0.55% respectively. The decays for the bending modes in the two directions are shown in Fig. 6(a) and Fig. 7(a), respectively. The power spectrum of the decay responses in the longitudinal and lateral directions and identified natural frequencies are shown in Fig. 6(b) and Fig. 7(b), respectively.

3. Wind tunnel tests

3.1. Overview of wind tunnel test

The inflow directions with yaw angles of $\beta = 0^\circ$ and $\beta = 90^\circ$ are in X and Z directions respectively, as shown in Fig. 8. The locations of the displacement measurements and the installation of aeroelastic model are the same in the dynamic characteristic tests, as previously shown in Figs. 4 and 5. The incoming velocity was measured using an ultrasonic anemometer named Gill Wind Master Pro with a sampling frequency of 500 Hz and measurement duration of 60s. The ultrasonic anemometer

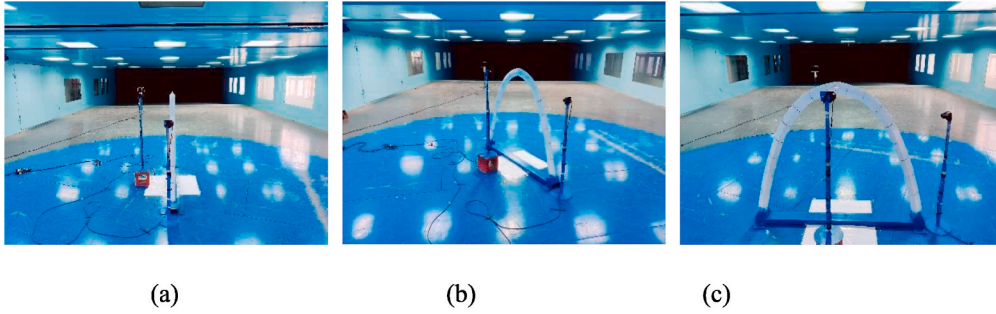


Fig. 9. Aeroelastic model in wind tunnel at different yaw angles: (a) $\beta = 0^\circ$; (b) $\beta = 35^\circ$; (c) $\beta = 90^\circ$.

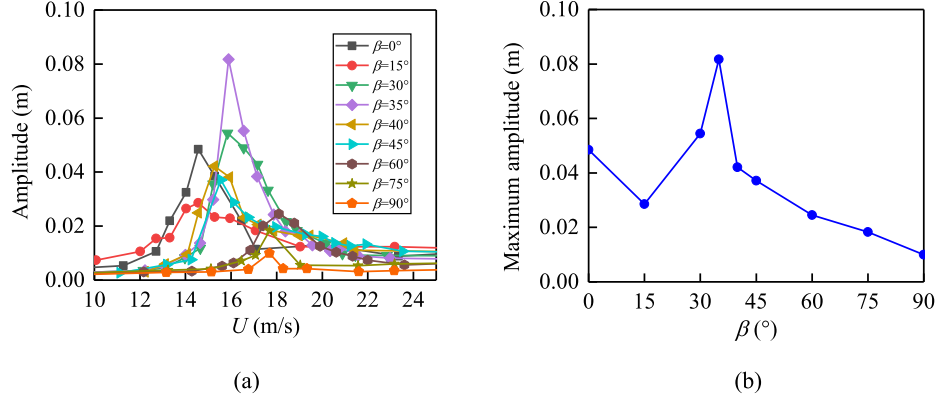


Fig. 10. Longitudinal VIV amplitudes of the arch tower in smooth flow (full-scale value): (a) amplitude versus wind velocity for different yaw angles; (b) maximum amplitudes versus yaw angles.

was located at the side of the aeroelastic model, so the model and the anemometer do not interfere with each other. A laser displacement meter was placed in the downstream of the aeroelastic model and arranged in the lateral direction. The turntable can be rotated clockwise to ensure that the laser displacement meter is always in the downstream of the aeroelastic model, avoiding possible significant interference with the flow field behind the aeroelastic model. The yaw angles of the inflow wind were realized by rotating the aeroelastic model, as shown in Fig. 9. The yaw angles $\beta = 0^\circ$, $\beta = 35^\circ$ and $\beta = 90^\circ$ are plotted in Fig. 9(a), (b) and (c) respectively.

3.2. VIV performance of the arch steel bridge tower

Wind tunnel tests were carried out on the aeroelastic model at different yaw angles ($\beta = 0^\circ, 15^\circ, 30^\circ, 35^\circ, 40^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ$) of wind

inflows. The RMS (root mean square) value of vibration in each direction is taken as the vibration amplitude. For different yaw angles, the variation of longitudinal and lateral vibration amplitudes of the model with wind velocity is shown in Figs. 10 and 11 respectively. According to the test results, VIV happened in the longitudinal direction of the tower for a few yaw angle of wind flow, and a most significant VIV phenomenon was observed with a yaw angle of 35° . Fig. 10(b) shows the variation of the maximum amplitudes with yaw angles. Due to irregularity of aerodynamic shape of the tower cross-sections, it is seen that the VIV amplitude of the tower varies significantly with yaw angles of wind flow. The maximum amplitude is 0.082 m at 35° yaw angle. The bridge tower has a special box section which is a blunt body section, leading to an unfavorable aerodynamic performance. Therefore, this bridge tower is prone to VIV phenomenon.

According to previous relevant research on VIV of tower structures,

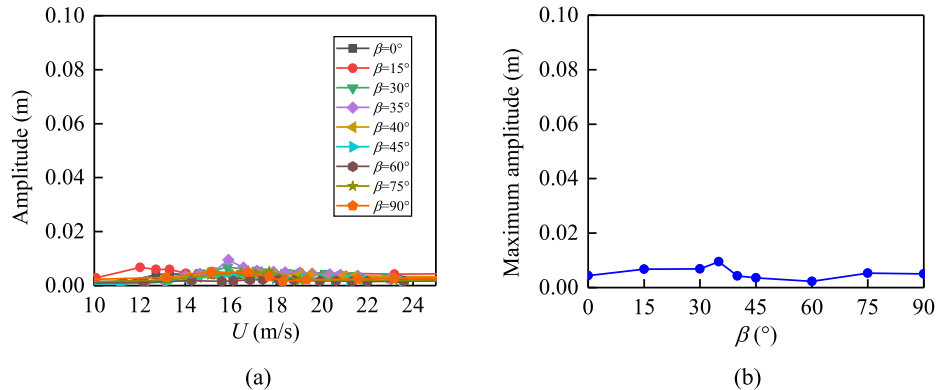


Fig. 11. Lateral vibration amplitudes of the arch tower in smooth flow (full-scale value): (a) amplitude versus wind velocity for different yaw angles; (b) maximum amplitude versus yaw angles.

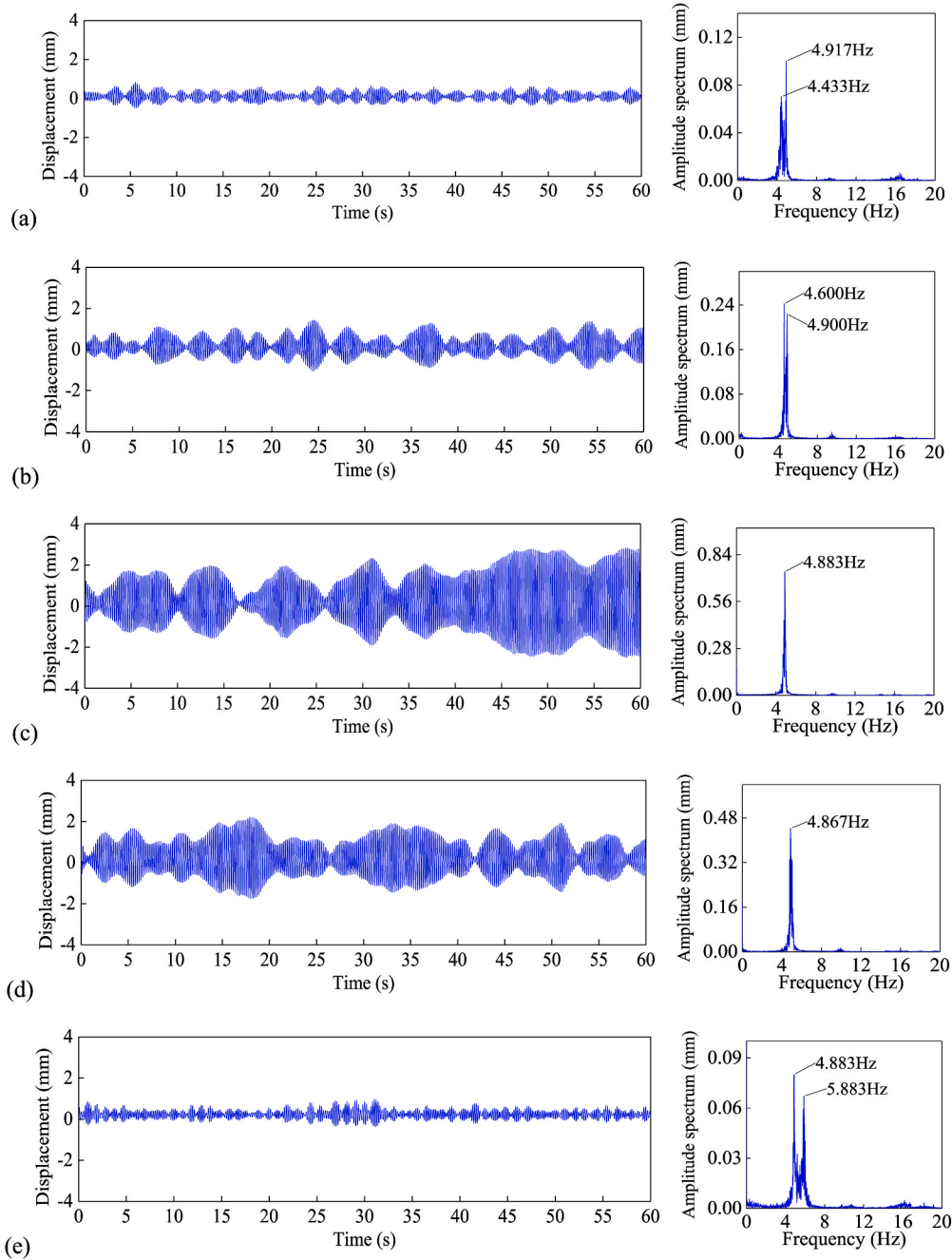


Fig. 12. Longitudinal displacement and its power spectrum of the aeroelastic model for $\beta = 35^\circ$: (a) 2.07 m/s; (b) 2.15 m/s; (c) 2.25 m/s; (d) 2.34 m/s; (e) 2.77 m/s.

the maximum amplitude usually corresponds to a yaw angle of 0° (Zhu et al., 2011; Marra et al., 2017; Li et al., 2021). In this study, the reason why the maximum VIV amplitude occurs with the yaw angle of 35° is that this bridge tower structure is a five-sided box type. When the yaw angle is 0° , the VIV amplitudes were suppressed with improved aerodynamic performance of the bridge tower structure since the bridge tower section is similar to an air nozzle section in the direction of incoming flow. Fig. 11 shows that there is no obvious VIV phenomenon in the lateral direction, mainly because the natural frequency of this mode is high, and the vortex shedding frequency is far away from the resonant frequency of the bridge tower for low wind speed.

Time domain and frequency domain responses for the model-scale longitudinal displacements are shown in Fig. 12 for $U = 2.07\text{--}2.77$ m/s with the yaw angle of 35° . It is observed that the beating phenomenon appears in displacement responses. A possible explanation for this

beating effect is that the fluctuations of the mean wind may cause lock-out from the lock-in range and eventually return into it (Lupi et al., 2021). Moreover, the possible shear flow and the variable cross-section may make the beating more pronounced. Fig. 12(a) shows the responses when the wind velocity is 2.07 m/s, in which the vibration amplitude is relatively small and there is no single resonance peak in the spectrum. In Fig. 12(b), the amplitude begins to increase, and a single peak tends to appear around the resonance frequency in the spectrum. The maximum VIV amplitude is reached with a wind speed of 2.25 m/s, as shown in Fig. 12(c). In the case, a single resonance peak appears in the spectrum, and the significant VIV is accompanied with a considerable beat vibration. As shown in Fig. 12(d), the amplitude begins to decrease, and the resonance peak tends to separate in the spectrum. As shown in Fig. 12(e), when the wind speed is 2.77 m/s the vibration amplitude drops significantly, and a significant separation for the resonance peaks can be

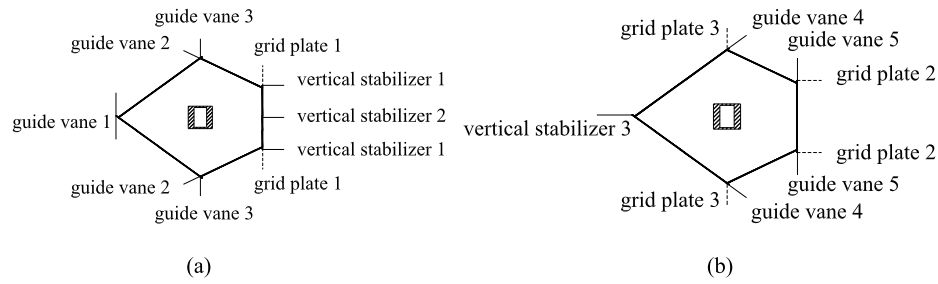


Fig. 13. Typical aerodynamic measures: (a) Group A; (b) Group B.

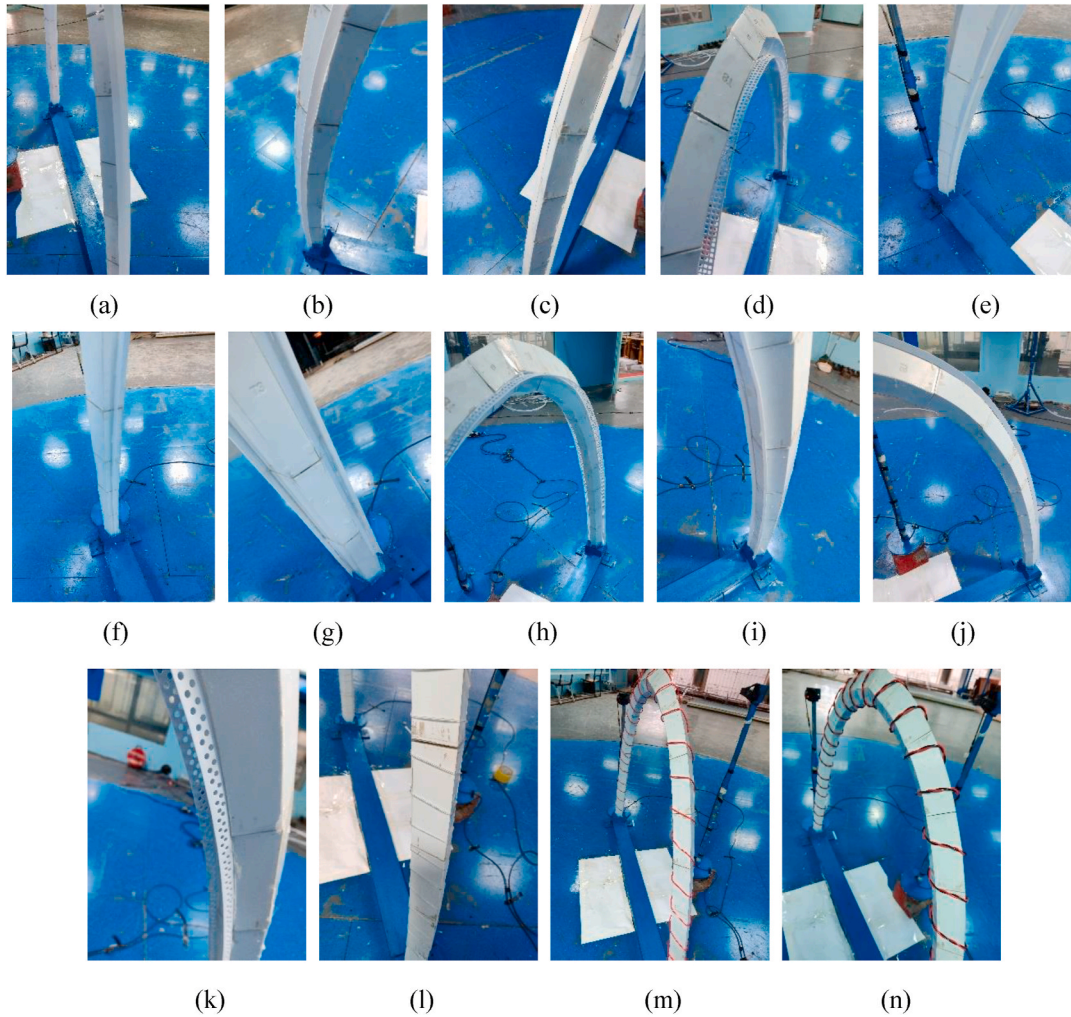


Fig. 14. Types of aerodynamic measures: (a) guide vane 1; (b) guide vane 2; (c) guide vane 3; (d) grid plate 1; (e) vertical stabilizer 1; (f) vertical stabilizer 2; (g) guide vane 4; (h) grid plate 3; (i) guide vane 5; (j) vertical stabilizer 3; (k) grid plate 2; (l) helix 1; (m) helix 2; (n) helix 3.

observed.

3.3. Aerodynamic measures for VIV mitigation

According to the previous experience of VIV mitigation measures, the VIV could be suppressed through a number of aerodynamic measures (Zhou et al., 2017; Zhu et al., 2011; Wang et al., 2021). The aerodynamic measures in this study are mainly aimed at optimizing the aerodynamic performance of the bridge tower by installing a stabilizer, guide vane or grid plate at the corner of the box section where vortex shedding is easy to occur. Different types of aerodynamic measures were designed for the bridge tower section, as shown in Fig. 13. As shown in

Fig. 13, two groups of aerodynamic measures are generally provided. The aerodynamic measures of Group A in Fig. 13(a) generally have positive effect for VIV mitigation; while those of Group B given in Fig. 13(b) were proved to have negligible mitigation effect or even resulted in worse effect. Fig. 14 shows the installations of various aerodynamic measures including guide vanes, grid plates, stabilizers and helix lines on the aeroelastic model.

Wind tunnel tests were carried out to study the VIV mitigation effect of above aerodynamic measures for 35° yaw angle of wind inflow. Again, the root mean square (RMS) of the measured displacement is scaled to represent the vibration amplitude of the full-scale bridge tower. The definitions of different aerodynamic measures and the

Table 6Effects of aerodynamic measures on VIV mitigation for $\beta = 35^\circ$ (full-scale).

Measure	Notes of measure	VIV amplitude (m)	Reduction ratio (%)
c0	original section	0.082	–
c1	guide vane 1 (width 1.75m)	0.048	41
c2	guide vane 2 (width 0.75m)	0.062	24
c3	guide vane 3 (width 0.75m)	0.043	48
c4	grid plate 1 (width 0.75m)	0.052	37
c5	vertical stabilizer 1 (width 0.75m)	0.026	68
c6	vertical stabilizer 2 (width 0.75m)	0.059	28
c51	vertical stabilizer 1 (width 0.25m)	0.062	24
c52	vertical stabilizer 1 (width 0.50m)	0.026	68
c53	vertical stabilizer 1 (width 1.00m)	0.050	38
c54	grid plate 2 (width 0.75m)	0.034	59
c7	helix 1 (diameter 0.1m, spacing 2.5m)	0.058	29
c71	helix 1 (diameter 0.1m, spacing 5.0m)	0.067	19
c72	helix 1 (diameter 0.1m, spacing 1.0m)	0.071	13
c73	helix 2 (diameter 0.15m, spacing 5.0m)	0.069	16
c74	helix 3 (diameter 0.30m, spacing 5.0m)	0.062	24
c11	guide vane 1 (width 1.00m)	0.085	–3
c21	guide vane 4 (width 0.75m)	0.080	2
c31	grid plate 3 (width 1.00m)	0.106	–29
c41	guide vane 5 (width 0.75m)	0.069	15
c61	vertical stabilizer 3 (width 0.75m)	0.097	–18

corresponding mitigation effects are summarized in Table 6, and the comparison of VIV mitigation effects of the aerodynamic measures is shown in Fig. 15. According to the test results, the aerodynamic measure with the best VIV mitigation effect is the vertical stabilizer 1 (width 0.75m), with the VIV magnitude reduced to 68% of that of the bridge tower with the original section.

Fig. 15(a) showed the variation of VIV amplitude with wind velocity for the tower with and without aerodynamic measures of Group A. It is seen that there is no peak in amplitude curve of VIV when the 'c5' aerodynamic measure was employed. It is possible that the 'c5' aerodynamic measure could disturb the bridge tower section corner vortex. Thus, the development of the VIV is significantly suppressed, leading to a favorable VIV mitigation effect. The 'c5' aerodynamic measures are further studied with different width of vertical stabilizer, and the results are given in Fig. 15(b). The test results show that vertical stabilizer 1 with a full-scale width of 0.75m reaches the best effect. When the width of the vertical stabilizer exceeds 1.00m or is smaller than 0.25m, the VIV mitigation effect decreases significantly. When the width is 0.50m, the VIV mitigation ratio is 68%, indicating that the dimension parameters of the aerodynamic measures have a significant impact on the VIV mitigation effect.

As shown in Fig. 15(c), the VIV mitigation effects of helix with different diameters and spacing is compared and analyzed. The results show that helix 1 (diameter 0.1m, spacing 2.5m) has the best VIV mitigation effect when only installing helix wires, but the VIV mitigation rate is only 29%. In Fig. 15(d), the effect of aerodynamic measures of Group B are compared and analyzed. It is found grid plate 3 (width 1.00m) not only fails to reduce vibration but also amplifies the amplitude by 29%.

According to the above studies, vertical stabilizer 1 with a width of 0.75m has the best VIV mitigation effect at the yaw angle of 35° . This

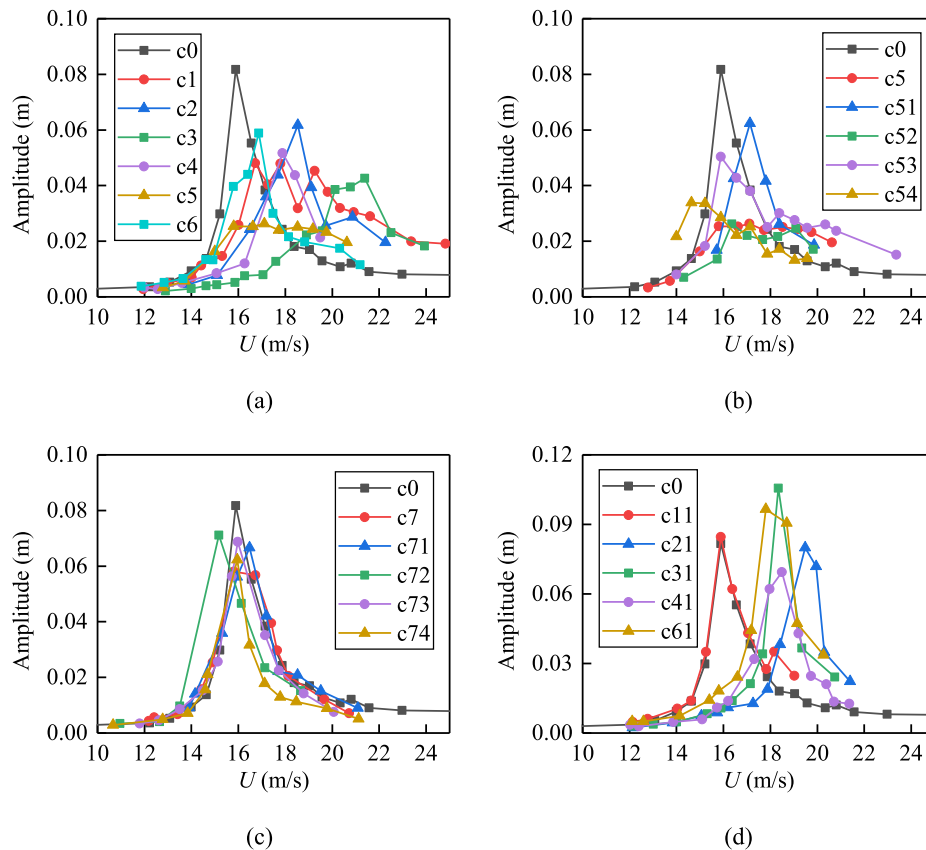


Fig. 15. Longitudinal VIV amplitude of tower with various aerodynamic measures for $\beta = 35^\circ$ (full-scale): (a) Group A; (b) size and ventilation effects of c5; (c) helix; (d) Group B.

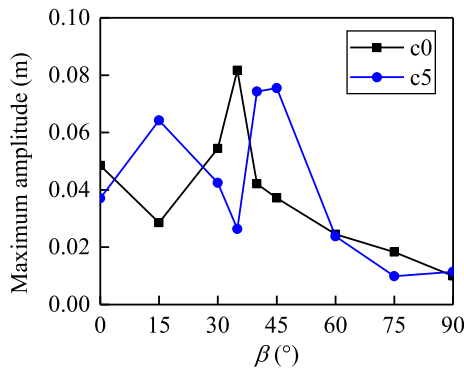


Fig. 16. Maximum VIV amplitudes versus yaw angles for aerodynamic measures of c_0 and c_5 .

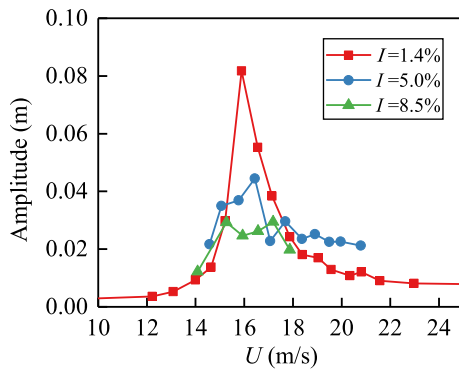


Fig. 17. Longitudinal VIV amplitude of the tower versus wind velocity under different turbulence intensities for $\beta = 35^\circ$ (full-scale).

kind of aerodynamic measures were further tested at different yaw angles of wind flow, and the results were shown in Fig. 16. It is seen that VIV amplitude of 0.076m was observed at the yaw angle of 45° , and the VIV amplitudes are 0.064m and 0.074m when yaw angles are 15° and 40° respectively. The VIV amplitude is amplified for some yaw angles when adopting the aerodynamic measure of vertical stabilizer 1. Due to the complicated geometric shape of the cross-section, it is difficult to find suitable aerodynamic measures for all yaw angles.

3.4. Effect of turbulence intensity

The VIV response of the bridge towers can be significantly influenced by the wind turbulence which has an effect on the vortex shedding of bridge tower sections. Thus, in this study, the effect of wind turbulence on the VIV amplitudes of the arch steel bridge tower was studied. The longitudinal VIV amplitudes of the tower under different turbulence intensities were analyzed. The turbulent flow was generated by several identical triangular spiers located in front of the model. The triangular spiers were aligned in the same cross section of the wind tunnel which was about 10 m far from the model. During the test, turbulence intensity was varied by changing the number of the triangular spiers. In this way, the flow in the same height kept uniform after adequate development, and a wind profile was generated to simulate the real condition in the field. The nominal turbulence intensity was measured by the ultrasonic anemometer, which had a constant location in this study. Finally, the nominal turbulence intensity reached 5.0% and 8.5% when 3 and 5 triangular spiers were employed, respectively. Fig. 17 compared the VIV results of the tower for $\beta = 35^\circ$ under different turbulence intensities. According to the data, when the turbulence intensity increases from 1.4% to 5.0%, the maximum VIV amplitude drops by 46%. When the turbulence intensity increases from 1.4% to 8.5%, the maximum VIV

Table 7

Geometric parameters of the eddy current damper (unit: mm).

Geometric parameter of component	Value
Magnet hollow disk ($D_1 \times d_1 \times \delta_1$)	$15 \times 5 \times 3$
Iron hollow disk ($D_2 \times d_2 \times \delta_2$)	$15 \times 5 \times 3$
Air gap	1
Copper conductor tube ($D_3 \times d_3 \times h_3$)	$19 \times 17 \times 70$
Iron magnetic conductive tube ($D_4 \times d_4 \times h_4$)	$21 \times 19 \times 70$
Aluminum guide rod ($D_5 \times h_5$)	5×120

amplitude drops by 65%, indicating that the turbulence intensity has an obvious effect on the VIV amplitude of the tower.

3.5. Effect of structural damping

In general, increasing the structural damping is a favorable measure for VIV mitigation. There are many ways to increase damping ratios, including viscous dampers, tuned mass dampers and tuned liquid column dampers. For aeroelastic model in wind tunnel, there is a lack of convenient and quick ways to increase the damping due to the difficulty to design the tiny dampers. In this study, a sleeve-type eddy current damper for aeroelastic model test is designed based on the permanent magnet eddy current damping technology (Chen et al., 2020). The configuration of the proposed damper is simple with as its manufacturing and assembling is convenient during the tests. It can effectively provide adjustable additional damping to the aeroelastic model by changing the number of magnets. Moreover, it is a small size device with sufficient damping coefficient, and it is effective for the multi-mode vibration mitigation as a reserved measure.

Geometric parameters of the eddy current damper are given in Table 7. D , d and h represent the outer diameter, inner diameter, thickness and height respectively, while the subscript “1”, “2”, “3” and “5” for D , d , δ and h denote the magnet hollow disk, iron hollow disk, copper conductor tube, iron magnetic conductive tube and aluminum guide rod respectively. Details of the eddy current damper are shown in Fig. 18. The damping coefficients of the eddy current damper with 2 magnets and 4 magnets are 7.4N-s/m and 13.6N-s/m respectively, which was measured from a simplified single-degree-of-freedom frame (details not given). The aluminum guide rod of the damper is connected with the aeroelastic model through the hot melt adhesive, and the outer sleeve is fixed on the cross-bar of column. The eddy current damper installed in the aeroelastic model is shown in Fig. 18(b).

The damping ratios were calculated using the logarithmic damping decrement method on the data obtained from the free decay tests of the free-standing tower model, as shown in Fig. 19. When the two and four magnets are installed, the damping ratios of the aeroelastic model for the bending mode in longitudinal direction are 0.63% and 0.86% respectively, as shown in Fig. 19(a) and 19(b). It is noted that the modal damping ratio of the aeroelastic model reaches 1.79% when the damper is installed into the higher position (the middle of T2 segment, detailed result is not given). However it may affect the appearance of the bridge and is not allowed by the bridge designer. Rather than increasing installation height, another way was adopted to increase the damping coefficient of the eddy current dampers. For illustration, Fig. 20 shows the variation of the modal damping ratios of the first mode with damping coefficient in full scale, which is obtained by complex eigenvalue analysis. It is confirmed that the damping ratio can be increased to target value by increasing the damping coefficients.

The longitudinal VIV amplitudes with various wind velocities under different damping ratios for $\beta = 35^\circ$ are shown in Fig. 21. The VIV mitigation rate reaches 38% when the damping ratio increases from 0.30% to 0.86%, indicating that the proposed damper is effective for the VIV mitigation of the tower by increasing the damping ratio. However, VIV mitigation rate is only 13.4% when the damping ratio increased from 0.30% to 0.63%. It shows that the damping ratio of the arch steel

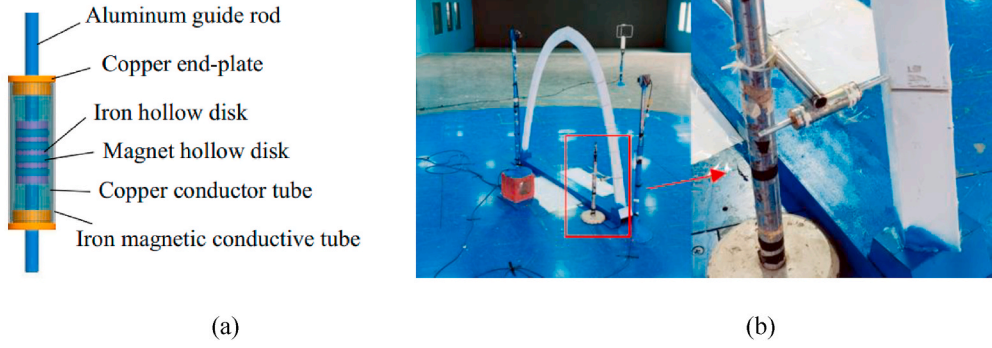


Fig. 18. Eddy current damper: (a) details; (b) installed in the aeroelastic model.

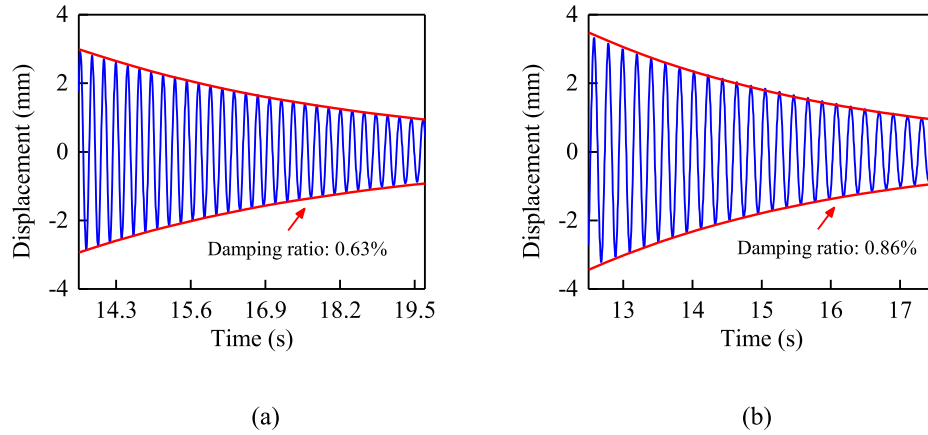


Fig. 19. Longitudinal displacements of the aeroelastic model with the eddy current damper from free decay tests: (a) two magnets; (b) four magnets.

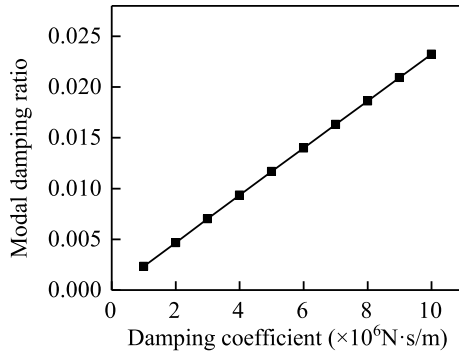


Fig. 20. Modal damping ratios with various damping coefficients in full-scale.

bridge tower is necessary to be added to a certain value more than 0.86% to suppress significantly the VIV amplitude. When the damping ratio reaches 1.79%, no VIV is observed. The variations of wind speed of the peak response with damping ratios are observed in Fig. 21. Two possible reasons might be responsible for the phenomenon. The first is that the variation of damping with amplitude is nonlinear. The second is that the aerodynamic force with various amplitudes and wind velocities has a highly nonlinear (Nguyen et al., 2018; Hu et al., 2018; Wang et al., 2022).

Increasing structural damping is chosen for reserved emergence measures in case that VIV of the free-standing tower happens. However during the free-standing state of the arch tower, no VIV is observed on site, possibly due to the high turbulence intensity. Auxiliary damping device was not applied to the bridge. After completion of the bridge, no VIV was found for the past two years. Illustration of the cable-stay bridge

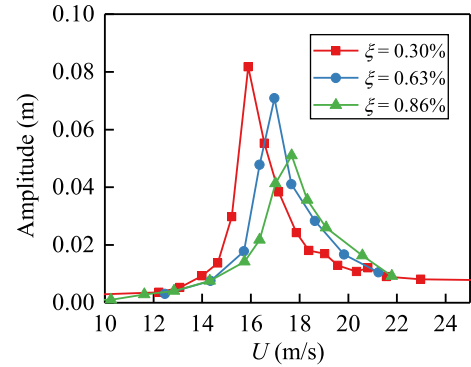


Fig. 21. Longitudinal VIV amplitude versus wind velocity for different damping ratios ($\beta = 35^\circ$).

is shown in Fig. 22.

4. Conclusions

This paper presents an experimental study on the VIV performance of an arch steel bridge tower through wind tunnel tests by employing an aeroelastic model. The VIV performances of the tower with various yaw angles are studied in smooth flow. A number of aerodynamic measures are taken to suppress the VIV amplitude in the most unfavorable yaw angle ($\beta = 35^\circ$). The effect of turbulence intensity on the VIV performances of the tower is evaluated. A sleeve-type eddy current damper installed in the aeroelastic model is proposed to increase the damping as a reserved measure, and the VIV performances of the tower with various



Fig. 22. Illustration of the cable-stayed bridge: (a) free-standing arch tower; (b) bridge completion.

damping ratios are investigated. Main conclusions are presented as follows.

- (1) VIV of the steel arch tower occurs for the fundamental longitudinal mode at several yaw angles, and the maximum VIV amplitude reaches 0.082m at the 35° yaw angle. The VIV phenomenon of the first-order lateral bending mode was not observed. The VIV amplitude drops by 50% when a turbulence intensity of 8.5% is applied.
- (2) Good VIV mitigation effects were observed when the aeroelastic model of arch steel bridge tower is with several aerodynamic measures at the 35° yaw angle, including vertical stabilizer 1 (width 0.75m), guide vane 1 (width 1.75m), guide vane 3 (width 0.75m) and grid plate 1 (width 0.75m). Their VIV mitigation rates are 68%, 41%, 48% and 37% respectively. Meanwhile, not only the geometric profiles of the aerodynamic measures, but also their width have obvious effects on the VIV mitigation of the tower. However, some of these aerodynamic measures do not show positive effects of VIV mitigation for other yaw angles.
- (3) Increasing structural damping is chosen for reserved measures in case that VIV of the free-standing tower indeed happens. The proposed sleeve-type eddy current damper is able to provide additional damping in the aeroelastic model. The modal damping ratios of the aeroelastic model reach 0.63% and 0.86% respectively when two and four magnets are installed in the damper, compared with the original modal damping ratio of 0.30%. For the actual bridge, VIV was not observed during the free-standing state of arch tower, and damping device is not actually applied to the bridge. After completion of the bridge, no VIV was found for the past two years.

CRediT authorship contribution statement

Yangfan Shen: Investigation, Methodology, Formal analysis, Writing – original draft. **Chaoqun Wang:** Writing – review & editing. **Aiguo Yan:** Supervision, Conceptualization. **Chao Chen:** Investigation, Supervision, Writing – review & editing. **Ou Yang:** Conceptualization, Funding acquisition. **Xugang Hua:** Supervision, Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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